

CHAPTER 11

The Economics of Immigration

Answers to End-of Chapter Questions

1. Does not apply.
2. (LO2) The point system was implemented in 1967 in order to make the selection criteria more objective. Before that time, applicants from the US and Commonwealth countries, such as the United Kingdom, were heavily favoured over applicants from many developing countries. It resembled a “buddy system,” and since most of the native Canadians were either aboriginal or of European descent, whites were heavily favoured. The power of selection was supposed to be less subject to the preferences of government bureaucrats and more subject to transparent criteria that could be evaluated. To the extent that these criteria are applied, applicants are to be assessed according to their ability to succeed in the Canadian labour market. Ideally, the policy seeks to fill positions for which there is a shortage of qualified Canadian candidates. As the text mentions, such a policy is effective only for the assessed classes of immigrants. With the family class growing in importance, these criteria have less of an impact on the skills of immigrants in general.

The empirical evidence is discussed based on Figure 11.1 and Table 11.1. The main point is the importance of the source country. The point system, when it is applied to the assessed class, does have the effect of screening out a lot of applicants from certain countries that tend to have low-skilled applicants. It does not appear to favour the most skilled applicants from any given country, which is the explicit aim of the policy. In other words, if applicants were to be sorted according to their country of origin, one would have approximately the same outcome as the point system, which categorizes them according to the criteria described in Exhibit 11.1 of the text.

3. (LO4) Economics assimilation is explained in Figure 11.5 of the text. It means essentially that after controlling for the standard, observable variables which tend to influence earnings, such as education and experience, the earnings levels of immigrants reach parity with those of native-born Canadians. In other words, it is wage gains to experience spent in the host country above and beyond the gains to experience observed for natives. The interpretation of this economic phenomenon is that immigrants are expected to enter the Canadian labour market with some disadvantages compared to native-born Canadians, as reflected in the so-called entry effect. This might be thought of as the entry disadvantage or penalty. It is akin to competing in a race with a starting line well behind your rivals. As time passes, however, immigrants may experience greater returns to experience than do native-born Canadians, permitting them to catch up in terms of earnings.

The problems that arise when using a single cross-section of data to estimate assimilation profiles are similar in spirit to those outlined in Chapter 4 in the context of disentangling age from birth-year effect. This is illustrated on Figure 11.6 of the text. It is necessary to understand what is meant by the term “cohort.” A cohort is an unchanging group of workers (i.e., the same people) that is followed over time as they age. The baby boom is an example. Baby boomers were born between 1946 and 1964, which means that in 2006, this generation is between 42 and 60 years old. The labour market outcomes of this cohort can then be compared to another cohort, such as the so-called Generation X. The underlying problem is certain unobservable factors which determine earnings. In order to accurately assess the assimilation effect, one needs to evaluate both the entry effect (how far behind are immigrants when they first arrive?) and the rate of wage growth (how much faster, if at all, do immigrants' earnings grow over time?). Then we can combine the starting point with the rate of wage growth to determine if the immigrants caught up to their native counterparts at the finish line. That is the essence of assimilation. Different cohorts of immigrants have different rates of assimilation. More specifically, there is evidence to believe that the “quality” of immigrant has deteriorated, on average, in both the United States and Canada. This refers to immigrants’ ability to realize labour market outcomes on a par with native-born workers. As the average skill and education level of immigrants falls over time relative to their native counterparts, the entry effect will increase. This implies that more recent immigrants (for example, those who came in 1995) will be starting their work careers at a lower level of earnings than those who immigrated earlier (for example, those who came in 1990), even after adjusting for inflation. If one compares the earnings of the first group in 1995 to those of the second group in 1995 (the second group has spent five more years in Canada), this figure will overestimate assimilation because there are two factors determining the difference in the immigrants’ earnings levels in 1995: (1) the growth in earnings and (2) the entry effect. Recall that the latest cohort of immigrants started at a lower level of earnings than did the preceding cohort. The first step in addressing this problem is to have data on the same cohort of immigrants at several points in time. The quasi-panel approach that is mentioned in the textbook is somewhat difficult to grasp.

Many, but not all, economists believe that the immigrant selection process should be improved. The criteria could be weighted more heavily toward immigrants’ suitability for the Canadian labour market. An even more controversial recommendation advanced by some is that the level of immigration should be reduced, or at least weighted more heavily to the assessed classes. It is admittedly somewhat difficult to gauge how adaptable an immigrant is to the Canadian labour market.

4. (LO3) This oversimplified theoretical analysis appears in panel (a) of Figure 11.4 of the text. A slightly more sophisticated analysis, which includes the impact on labour demand, appears in panel (b). This analysis is within the framework of elementary supply-and-demand analysis. Immigration activity should definitely expand the supply of labour, and it might have an indirect effect of slightly expanding the demand for labour. A crucial, and often overlooked, variable is the degree of complementarity between the labour of

natives and that of immigrants. In some cases, immigrants fill vacancies in the labour market, both in very highly skilled jobs and in modestly skilled jobs. To the degree that their services are complements rather than substitutes for native labour, immigration is economically beneficial.

Empirically, there is some evidence that suggests that this statement might be true, but not to a great extent. Other evidence suggests that immigration does not adversely affect native-born low-skilled workers at all. A consensus has not been reached. There are at least two complications to surmount in estimating the impact of immigration. First, there is the problem of establishing the counterfactual. We simply cannot observe whether (or by how much) low-skilled workers would earn more — or have better job opportunities — in the absence of immigration. For example, almost all taxi drivers in major Canadian cities are immigrants. One cannot observe the case in which all taxi drivers are born in Canada. It is difficult to determine whether wages would be higher in their absence, or high enough to induce native-born Canadians to enter the occupation. The other problem is associated with simultaneity, or the joint determination of migration decisions and the effect of immigration on the labour market. Immigrants tend to choose destinations where labour markets are relatively strong. This implies that if there is an adverse impact in certain labour markets, the impact will be understated because the two effects work in opposite directions. Perhaps immigration displaces workers, worsening labour market conditions, but the immigrants will not come to a region at all unless the labour market is relatively healthy. It is difficult to isolate these two effects, which may both be generating the outcomes that are observed.

5. (LO3) One has to compare outcomes across cities in order to analyze the phenomenon, so these data do hold some promise. A totally fallacious analysis would proceed as follows. One would correlate the percentage of foreign-born workers as a share of the labour force in each area with the average wage level prevailing in each area. If the correlation is negative, one would conclude, erroneously, that immigration has adverse effects on native-born workers. This faulty analysis ignores the *ceteris paribus* condition. One has to account for as many of the other factors which affect labour market outcomes in these cities as possible. More specifically, the nature of the problem is the simultaneity, or joint determination, of immigration flows and labour market conditions. Immigration flows and labour market conditions influence each other in a two-way causal fashion. We seek to evaluate the magnitude of a one-way effect: how immigration flows affect the equilibrium wage levels. The difference-in-differences approach explained in the textbook is more appropriate, but still has its shortcomings. The best way to proceed is to attempt to analyze an instance in which a wave of immigration was determined by political forces rather than by economic forces. If immigration is driven by factors unrelated to the labour market conditions, then we know that it is the immigration activity that may affect the labour market conditions rather than the other way around. See page 320.

There is another approach, favoured by Borjas, Freeman, and Katz, discussed in the text,

in which the unit of analysis is not the geographical unit (CMA) but rather a particular grade of labour. The grades of labour are categorized by gender, age group, and education level. The strategy is to investigate whether a push of immigrants characterized by a given gender, age group, and education level tends to depress the average wages of their native counterparts. This approach would not use data like that presented in Figure 11.2 of the text.

6. (LO4) The reasons for this troubling phenomenon are not fully understood. It is thought that the labour market conditions that prevail at the time of entry have some effect. Those immigrants are likely to be very sensitive to the business cycle, and more so than the stock of native-born Canadians. This development is all the more surprising in light of the fact that immigrants since the 1990s have been pretty well-educated (more so than their native-born counterparts), which is a result of the points system. They thus tend to have high values for observed skills. The research has demonstrated that immigrants receive very little remuneration for the experience that they obtained in their former countries. That experience is simply not recognized by Canadian employers. It is thought that the decline in entry earnings of immigrants can be partly attributed to changes in the source countries of immigrants. Many immigrants do not arrive with strong language skills in English or French. It is possible that many of them worked in environments and cultures that are very different from that in which Canadian labour markets operate, making them less desirable to Canadian firms until they have adjusted to our professional milieu, which can take a long time.
7. (LO5) This is discussed at the end of the chapter. Although the text makes the point that this phenomenon affects developing countries the most, it also affects Canada to some degree. This is most visible in the health care and ICT industries. Generally, the education of doctors and software engineers at Canadian universities is very highly subsidized and significantly less costly to the student than is the case in the United States. For a variety of reasons, doctors, nurses, engineers, finance specialists, software professionals, and some university professors are much more highly paid in the United States than in Canada, and such professionals are thus tempted to emigrate from Canada. Reinforcing this trend is the higher level of taxation in Canada. It is difficult to combat this trend, which harms the Canadian health care system, the software industry, and ultimately the economy (because of the loss of very highly productive workers). Some analysts suggest that this is an argument for lower taxes in Canada. Ironically, a number of decades ago, Canada benefited enormously from a brain drain from the United Kingdom. A similar situation is also observed in the realm of professional sports, which one might call the “brawn drain.” It is driven by the same economic forces. Note that this “brain drain” from Canada to the US appears to have abated somewhat since 2000.
8. (LO5) The economic model of migration is still valid. All other factors held constant, one would expect the higher salaries paid in the US to attract many highly skilled Canadian workers to migrate, but there are all sorts of frictions that could occur that would work in the opposing direction. The greater the costs associated with migration, the higher the pay

differential has to be in order to induce migration. These countervailing costs can take the form of family ties and both implicit and explicit costs of moving. There is also the non-pecuniary factor of preferences. Some individuals have strong preferences for either a particular region in Canada or the Canadian way of life and culture. I would guess that all four of the authors of this textbook (who are based at Canadian universities) could have earned significantly higher salaries in the US. That they chose not to migrate supports the assertions that I have just made.

Answers to End-of Chapter Problems

1. (LO1, 4) There are two cohorts: immigrants who arrived during the period of 1991–95 and immigrants who arrived during the period of 1996–2000. They reflect two different groups of people. The entry effect for the first cohort cannot be evaluated from the table, as we do not know how much the native-born workers were earning at any time during the period of 1991–1995. The latest cohort of immigrants enters with a disadvantage (an entry effect is thought to be negative) of \$8,000 (\$40,000 – \$32,000). Note that between 2000 and 2005, the earnings of native-born Canadians grew slowly. This may be interpreted as the general growth of earnings in the economy, which should affect all groups (native and immigrant alike) in the same fashion. The 1991–1995 group of immigrants had a negative entry effect of \$8,000 in 2000 but reduced the gap relative to native-born Canadians to \$5,000 in 2005 (\$41,000 – \$36,000). This means that this cohort did assimilate somewhat because their earnings grew more quickly than those of native-born Canadians. They assimilated to the tune of \$3,000.

We cannot analyze the assimilation rates for the earlier cohort without making an assumption regarding their entry effect. To make an attempt, compare the earnings of the 1991–1995 cohort in 2000, after they have been in the country for five years (point B in Figure 11.6 of the text) to the earnings of later immigrants in 2000 (point D). If the initial earnings of the earlier cohort are the same (an entry effect of \$8,000 below the native-born), we can conclude that the earlier cohort assimilated to the tune of \$6,000 (\$38,000 – \$32,000). We know for certain that the earlier cohort did not assimilate at all between 2000 and 2005, as they gained no ground on the native-born Canadians. (Each group gained only \$1,000.)

The main point is that assimilation rates will typically vary across cohorts (here we have two cohorts, or waves, of immigrants), and/or across periods of time for the same cohort (e.g., the 1991–95 group between 1995 and 2000 and the same group between 2000 and 2005).

2. (LO3)
 - a) The gamma parameter measures the change in the unemployment rate, measured in percentage points, that stems from a 1% change in the inflow of immigrants. The expectation is that the sign will be ambiguous, as there are at least two opposing effects, as discussed on page 320 of the text. On the supply side of the

labour market, a force is acting that might make γ positive, but on the demand side, an effect in the opposite direction is expected.

- b) This estimated negative association likely reflects two-way causality, which can also be thought of as a form of simultaneity bias. The intended function of this regression equation is to model changes in the rate of unemployment as a function of immigration flows, but immigration flows in turn can be influenced by labour market conditions. It is difficult to disentangle those two effects with this data set and that particular econometric specification, so both effects are mixed into the generation of that estimate of -1.56. Another way of looking at this empirical problem is to point out that we do not observe the counterfactual equilibrium for the labour market, i.e. what the unemployment rate would be had there been a different level of immigration. The researcher cannot control independently the inflow of immigration: instead, the level of immigration inflow is determined by several factors, including the aggregate unemployment rate.

The researcher would be correct if it could be ensured that the immigration flows being measured are exogenous, meaning that they are determined independently of the national unemployment rate. The current consensus is that the economic interests of native workers who are low-skilled might be harmed (through lower wages than what would otherwise be the case) by significant inflows of immigrants. Among the works cited in the textbook, Card (2001) and Altonji & Card (1991) are the most relevant, along with any study that employs the “natural experiment” approach.

3. (LO4) Note that there was a global effect of very slow growth in earnings between 2005 and 2010 that affected all cohorts of immigrants and native-born Canadians.
- a) Entry effects are easy to calculate. Working from one cross-section of data (corresponding to the year of entry), simply subtract the earnings of native Canadian to the earnings of the newly arrived immigrant cohort in the year 2005. This gives $\$64,778 - \$49,898 = \$14,880$.
- b) Working from one cross-section of data and comparing the earnings of the two cohorts that are separated by five years, we have $\$60,881 - \$54,478 = \$6,403$. What we are doing is comparing the average wage of this cohort to the average wage of the cohort that arrived five years earlier at one point in time: 2005. An erroneous interpretation of this figure is the effect on wages of the IM9094 cohort of working in Canada for five more years. Projecting this growth to the earnings of the 90–94 cohort in 2010, we would add \$6,403 to \$54,478 in order to obtain \$60,881. This is equivalent to projecting the distance from points B and D to point E in Figure 11.6a of the text.
- c) Over the period of 2005–2010, the wages of all workers were growing. We are told that the true earnings in 2005 for IM9094 were \$59,185, so our estimate is

off. In order to measure the wage growth that is attributable to assimilation, follow the path of the 90–94 cohort from 2005 to 2010: we obtain \$59,185 – \$54,478 = \$4,707. This corresponds to the movement from D to E in Figure 11.6. Due to the global trends, this cohort gained ground relative to native workers, whose earnings rose by only \$67,247 – \$64,778 = \$2,469 over the same period. IM9094 thus assimilated to the tune of about \$2,238 (\$4,707 – \$2,469). The lesson learned is that you have to consider (a) the entry effect, (b) the rate of growth of earnings of the immigrant cohort, and (c) the rate of growth of earnings of the native-born workers in order to fully assess assimilation.

4. (LO3)

- a) b is the elasticity of labour supply with respect to the population, while β is the elasticity of labour demand with respect to the population. The immediate effect of immigration activity is an increase in the population. The magnitude of γ depends on the net effect of immigrants arriving and natives departing on the population of region. In other words, one has to consider in-migration and out-migration.
- b) By setting the S equation equal to the D equation and solving for e^* and W^* algebraically, one obtains the following expressions:

$$W^* = \frac{(a - \alpha) + P(b - \beta)}{\eta - c}$$

$$e^* = \frac{\eta a - \alpha c + (b\eta - c\beta)P}{\eta - c}$$

- c) Differentiate the two equilibrium conditions with respect to the variable P in order to obtain the following expressions:

$$\frac{\partial W^*}{\partial P} = \frac{(b - \beta)}{\eta - c}$$

$$\frac{\partial e^*}{\partial P} = \frac{(b\eta - c\beta)}{\eta - c}$$

In case (i), by substituting the numbers in, the values are $-1/0.4 = -2.5$ for the wage elasticity, and $-0.3/-0.4 = 3/4$ for the employment elasticity. With this set of values, the immigration inflow has a negative (and highly elastic) effect on wages, but a positive and inelastic effect on employment. There is a strong supply shift ($b = 1$) and no offsetting

demand shift at all ($\beta = 0$). In case (ii), by substituting the numbers in, the values are 0 for the wage elasticity and $-0.4/-0.4 = 1$ for the employment elasticity. With this set of values, the immigration inflow has no effect on wages, but a positive and unitary elastic effect on employment. There is a strong positive supply shift ($b = 1$) with a commensurate strong positive demand shift ($\beta = 1$) that fully offsets the decline in the wage that was obtained in part (i). Since both shifts are to the right, equilibrium employment must rise.

5. (LO4)

- a) In order to interpret the constant term α , take the exponential of -0.25 in order to generate the anti-log. This corresponds to the gap between the average wage of a cohort of immigrants and their native counterparts when the immigrants have just arrived; thus, $Y_{SM} = 0$. This gap, in turn, is expressed as a ratio of the earnings of immigrants to the earnings of natives. This is the entry effect, which should be negative, so that when they arrive in Canada, immigrants earn less than otherwise similar natives. In order for this gap to be zero, Y_{jt} -tilde has to be unity, which means that the ratio of the two wage levels is unity. The interpretation of β is the annual rate of catching up. Since the wages are expressed in terms of logarithms, each year worked in Canada since immigration corresponds to a narrowing of 1% in the wage gap that existed at the time of entry. Substituting the given values into that condition, we have $0.25 = 0.01 * Y_{SM}$, or $Y_{SM} = 25$. This implies that the assimilation process takes 25 years.
- b) This would imply that each cohort of immigrants has its own assimilation profile. This specification allows for some cohorts to assimilate faster than others. For each particular cohort, one would solve the equation $\alpha_j = \beta_j * Y_{SM}$. This value of Y_{SM} would correspond to the number of years that cohort j takes to catch up to its native counterpart.

6. (LO5) The easiest way to address this problem is to calculate the discounted present value of the discrepancy in the value of the two jobs. We can take this step because the time horizons and the growth rate in earnings are the same in each case. The job in the US pays \$25,000 more, and this stream of payments is augmented by 6% each year for a period extending 40 years into the future. In the following year, for example, the discrepancy is \$ 26,500, while in the following year, it is \$ 28,090. Applying the formula for calculating DPV, the first three terms are $25,000 + 26,500/1.05 + 28,090/(1.05)^2$. If we add up all 40 terms, we would have the amount by which the lifetime earnings in the US exceed the lifetime earnings in Canada. We cannot use the approximation given by the formula for calculating the discounted present value of a stream of payments paid in perpetuity, which is equal to simply the payment divided by the interest rate, because the payment is not constant in each period.

Taking only pecuniary factors into consideration, the value of the Canadian job is the negative of this figure. Non-pecuniary factors, such as family ties, working conditions and geographical preference, may weigh in the other direction, however, encouraging this

person to remain in Canada. He/she may have a partner with a good career in Canada, which would constitute a pecuniary factor weighing against migration. There may also be very high mobility costs. If so, we would expect this person to move sooner rather than later in his/her career.